

Prakruti Sudarshan | Ph.D. student

CONTACT INFORMATION

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EDUCATION

**Max Planck Institute for Astronomy &
Ruprecht-Karls-Universität Heidelberg**, Germany
Ph.D. in Astronomy, IMPRS Fellow

October 2022 - present

- Advisor: Dr. Mario Flock
- Working thesis title: *Dust dynamics in radiative protoplanetary systems*

Eberhard Karls Universität Tübingen, Germany
M.Sc. in Astro and Particle Physics, Overall Grade: 1.1

September 2022

- Advisors: Late Prof. Dr. Wilhelm Kley, Dr. Christoph Schäfer
- Thesis: *Spirals and gravitoturbulence in 2D disks*, Thesis grade: 1.0

Christ University, Bangalore, India

B.Sc. *triple major* in Physics, Mathematics and Chemistry, GPA: 3.93/4.00

June 2020

- Research Education and Advancement Program (REAP) student in Astrophysics, Jawaharlal Nehru Planetarium, Bangalore.

RESEARCH INTERESTS

I have 5+ years of experience in **planet formation theory**, studying **dynamics of starlight heated protoplanetary systems** using multifluid dust and radiation hydro- simulations with the PLUTO code and synthetic observations with radiative transfer methods. My PhD focus is on specifically, testing the effects of stellar irradiation on disk geometry. I am involved in **ongoing collaborations** modeling **planetesimals with streaming instability, dust modeling methods, gravitational instability, and circumbinary disks**.

TECHNICAL SKILLS

- **Programming languages:** C, Python (*Advanced*); Rust, CUDA, C++ (*Basic*)
- **Astrophysical Codes:** PLUTO with radiation multifluid hydrodynamics, RADMC3D (*Advanced*); n-body code Rebound, IDEFIX (*Intermediate*)
- **Data visualization & other tools:** L^AT_EX, Origin Pro (*Advanced*); Matlab, JavaScript, Go (*Intermediate*);
- **High Performance Computing** with MPI/GPUs (*Advanced*)

RESEARCH STAYS

Planet Formation Group, CCA, Flatiron Institute, New York City

Jan – Feb 2026

Visiting Student, working on a radiation-methods comparison and the dust module in PLUTO, hosted by Dr. Thomas Pfeil and Prof. Phil Armitage.

Computational Physics Group, Universität Tübingen, Germany

July 2021 – March 2022

HiWi student, worked on a project of **how cooling influences circumbinary disks** with the GPU-PLUTO code with Dr. Anna Penzlin, Dr. Alex Ziampras, Prof. Dr. Wilhelm Kley and Prof. Dr. Richard Nelson, related paper published in A&A (664, A157)

PUBLICATIONS: MAJOR CONTRIBUTION

Sudarshan, P., Ziampras, A., Flock, M., et al. in prep, “A multi-species fluid dust module for the PLUTO code”

Kavipurapu, P., **Sudarshan, P.**, Flock, M., et al. in prep, “Vertical dust stratification in VSI-turbulent protoplanetary disks”

	GRAVITY Collaboration. Sanchez-Bermudez, J., Sudarshan, P. , et al., in prep, “The GRAVITY young stellar object survey: multiplicity of FS CMa revealed with infrared interferometry”	
	Baronett, S., et al. (incl. Sudarshan, P.), in review (ApJ), “A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains” [ADS]	
	Sudarshan, P. , Flock, M., et al. 2026, “Starlight-driven flared-staircase geometry in radiation hydrodynamics simulations of protoplanetary disks” [ADS]	
	Ziampras, A., Sudarshan, P. , et al., 2025, MNRAS, 536, 4, pp.3322-3337 “Dusty substructures induced by planets in ALMA disks: how dust growth and dynamics changes the picture” [ADS]	
	Sudarshan, P. , Penzlin, A. B. T., Ziampras, A., Kley, W., & Nelson, R. P., 2022, A&A, 664, A157 “How cooling influences circumbinary disks” [ADS]	
CO-AUTHORED PUBLICATIONS	Cecil, M., et al. (incl. Sudarshan, P.), in press, A&A, “The role of detailed gas and dust opacities in shaping the evolution of the inner disc edge subject to episodic accretion” [ADS]	
	Ueda, T., et al. 2024 (incl. Sudarshan, P.), Nature Astronomy 8, 1148-1158 (2024) “Support for fragile porous dust in a gravitationally self-regulated disk around IMLup” [ADS]	
TEACHING AND SUPERVISION	Summer project supervision of <i>Niranjan Ashok</i> (→ <i>Grad student at Alabama</i>) Topic: Temperature and radiation fields at gap edges in protoplanetary disks	Summer 2025
	Master thesis co-supervision of <i>Pundari Kavipurapu</i> , Heidelberg University Topic: Dust dynamics in VSI active disks with parameterized cooling laws	2023 – 2025
	Advanced Astronomy Lab Tutor , Heidelberg University <i>F36 Adaptive optics: Wavefront analysis using the Shack Hartmann Sensor</i>	Summer 2023
TALKS AND POSTERS	PLUTO symposium, MPIA, Heidelberg Germany (<i>Contributed Talk</i>)	June 2026
	3rd PFITS+ Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation (upcoming <i>Invited Talk</i>)	June 2026
	Star and Planet formation seminar, European Southern Observatory (ESO), Germany	May 2026
	Birnstiel Group Retreat, Fraueninsel, Chiemsee (<i>Invited Talk</i>)	April 2026
	Accretion-Ejection French online seminar series (<i>Invited Talk</i>)	March 2026
	SPF talk, University of Michigan, Ann Arbor	March 2026
	Disks and Astrochemistry group meeting, Harvard & Smithsonian CfA, Boston	March 2026
	Steward Observatory theory group meeting, Tucson, Arizona	March 2026
	Hydrodynamics and Dusty Turbulence in Protoplanetary Disks, Flatiron CCA Workshop, New York City (<i>Invited Talk</i>)	February 2026
	Planets and Origins group discussion, Columbia University, New York City	February 2026
	Planet formation group meeting, CCA Flatiron Institute, New York City	February 2026
	ICEPF Conference 2025, Shanghai, China (<i>Poster</i>)	December 2025
	Exoplanets and Planet Formation Mini-Workshop, Tsinghua DoA, Beijing	December 2025
	Exo-ELT, European Southern Observatory (ESO), Germany (<i>Contributed Talk</i>)	November 2025
	AG Planet Formation Splinter, Görlitz, Germany (<i>Contributed Talk</i>)	September 2025
	Partially Ionized Plasmas in Astrophysics, Bergen, Norway (<i>Contributed Talk</i>)	September 2025
	Multiplicity in young stars, NBI, Copenhagen (<i>Contributed Talk</i>)	August 2025
	Planet & Star Formation Coffee, MPIA	March 2025

	Astronomical Society of India Meeting, Rourkela, India (<i>Contributed Talk</i>)	February 2025
	IMPRS retreat, Mosbach, Germany	September 2024
	PLUTO symposium, Torino, Italy (<i>Contributed Talk</i>)	September 2024
	Gravity+ conference, Grenoble, France (<i>Poster</i>)	June 2024
	Theory Group seminar, IPAG Grenoble, France	June 2024
	DisktoPlanet Group meeting, MPIA, Germany	May 2024
	Astronomy Unit Group meeting, Queen Mary University London	April 2024
	Group seminar, Imperial College London	April 2024
	MHD flows in young circumstellar disks, Ringberg, Germany (<i>Poster</i>)	September 2023
	Theory group meeting, Charles University, Prague	May 2023
	Protostars and Planets VII, Kyoto, Japan (<i>Poster</i>)	April 2023
	DisktoPlanet Group meeting, MPIA, Germany	October 2022
	The inner regions of protoplanetary disks, Ringberg, Germany (<i>Poster</i>)	September 2022
	Formation and Evolution of Planetary systems conference, Tübingen (<i>Contributed Talk</i>)	August 2022
	European Astronomical Society Annual Meeting (S22 session), Valencia, Spain (<i>Contributed Talk</i>)	July 2022
HONORS AND AWARDS	International Max Planck Research School (IMPRS-HD) Fellowship	2022 – 2026
	SN Bhatt Memorial ICTS Fellowship, 1/10 selected All India (<i>cancelled due to Covid-19</i>)	2020
	Christ University Physics and Electronics Club Quiz Champion	2019
	Research Education and Advancement Program (REAP) scholarship	2018 – 2019
PROFESSIONAL SERVICE / OUTREACH	SOC , “Early career astronomers and their supporters” session, European Astronomical Society Annual Meeting (EAS) Lausanne, Switzerland	July 2026
	SOC , Planet formation and Exoplanets (PFE) MPIA Department Retreat, Löwenstein, Germany	May 2026
	Journal Club Speaker, MPIA Girls’ day outreach program	April 2026
	Volunteer & Session Chair , PyConDE and PyData, Darmstadt, Germany	April 2026
	Co-chair & SOC , “Early career astronomers and their supporters” session, European Astronomical Society Annual Meeting (EAS) Cork, Ireland	July 2025
	Co-chair & SOC , “Early career astronomers and their supporters” session, European Astronomical Society Annual Meeting (EAS) Padova, Italy	July 2024
	Disk2Planet Group mailing list organizer	2024 – 2025
	LOC , Ringberg meeting on “MHD flows in young circumstellar disks”	October 2023
	Volunteer, MPIA Tag der offenen Tür (Open day)	October 2023
	LOC , Formation and Evolution of Planetary systems conference, Tübingen	August 2022
	Mentored students from under-resourced backgrounds on STEM career pathways, Viveka Scholar Program, SVYM, Sargur Village, India	December 2023
	Mediator , Virtual PSYCHE exhibition season, Science Gallery Bengaluru	April – May 2022
OBSERVING TIME	30-inch telescope, Vainu Bappu Observatory, Kavalur, India <i>Time series spectra of the Sun</i>	2 nights, 2019
INDUSTRY EXPERIENCE	Bosch Sensortec GmbH, Reutlingen, Germany Working Student Intern, Worked on the analysis of MEMS-based Bosch sensor data and in-house code documentation using Python and Matlab.	April 2022 – July 2022